

Ans 1:- Recall,

$a \equiv b \pmod{n}$ means $n \mid (a-b)$.

(i) Reflexive:- For any integer a , $a-a=0$ & $n \mid 0$.

Therefore,

$$a \equiv a \pmod{n}$$

Thus, R is reflexive.

(ii) Symmetric:- Suppose, aRb that is, $a \equiv b \pmod{n}$

Then, $\exists k, l$ such that

$$a-b = nk, \quad b-c = nl.$$

(iii) Transitive:-

adding gives.

$$\begin{aligned} a-c &= (a-b) + (b-c) \\ &= nk + nl = n(k+l). \end{aligned}$$

so $n \mid (a-c)$ & hence $a \equiv c \pmod{n}$, i.e.

aRc . Thus R is transitive.

Therefore R is an equivalence relation on $\mathbb{Z}$.

Ans 2:-

In general, for a +ve integer m & an integer x with $0 \leq x < m$ the congruence class (or Equivalence, Residual class) of $x \pmod{m}$ is:-

$$[x]_m = \{k \in \mathbb{Z} : k \equiv x \pmod{m}\}$$

(1) There are 5 distinct classes represented by residues 0, 1, 2, 3, 4 :-

$$\{ \dots, -10, -5, 0, 5, 10, 15, \dots \}$$

$$\{ \dots, -9, -4, 1, 6, 11, 16, \dots \}$$

$$\{ \dots, -8, -3, 2, 7, \dots \}$$

$$\{ \dots, -7, -2, 3, 8, 13, 18, \dots \}$$

$$\{ \dots, -6, -1, 4, 9, 14, \dots \}$$

(2) There are 4 distinct classes, represented by residues 0, 1, 2, 3 :-

$$\{ \dots, -8, -4, 0, 4, 8, 12, \dots \}$$

$$\{ \dots, -7, -3, 1, 5, 9, 13, \dots \}$$

$$\{ \dots, -6, -2, 2, 6, 10, 14, \dots \}$$

$$\{ \dots, -5, -1, 3, 7, 11, 15, \dots \}$$

Ques 3 (i) let $\gcd(a, b) = 1$ & a/bc
 $\therefore \gcd(a, b) = 1$, by Bezout's identity
 $\exists$ integers x, y such that,
 $ax + by = 1$
 multiply both sides by c ,
 $acx + bcy = c$

Now as, a/bc . By hypothesis a/bc so a/bcy as well. Therefore both the terms on L.H.S are multiple of a , so their sum is multiple of a .
 Hence a/c .

(ii) Assume, $ac \equiv bc \pmod{m}$ & $\gcd(c, m) = 1$
 The congruence $ac \equiv bc \pmod{m}$ means,
 $m \mid (ac - bc) = c(a - b)$

Thus m divides $c(a - b)$.

Given that $\gcd(c, m) = 1$,
from part (i) with $a = m$, $b = c$ &
 c is replaced by $(a-b)$. The condition
 $\gcd(m, c) = 1$ holds & from $m/c(a-b)$ it
follows by part (i) that
 $m/(a-b)$

$$\therefore a \equiv b \pmod{m}.$$

H.P

4.

Q1-
$$x \equiv 3 \pmod{9}$$

$$x \equiv 7 \pmod{13}$$

$$x = (a_1 m_1 m_1^{-1} + a_2 m_2 m_2^{-1})$$

$$x \equiv a_1 \pmod{m_1} \quad | \quad x \equiv 3 \pmod{9}$$

$$x \equiv a_2 \pmod{m_2} \quad | \quad x \equiv 7 \pmod{13}$$

Given		to find		
$a_1 = 3$	$m_1 = 9$	m_1	m_1^{-1}	M
$a_2 = 7$	$m_2 = 13$	m_2	m_2^{-1}	

$$m = m_1 \times m_2$$

$$m = 9 \times 13 = 117$$

Now,

$$M_1 = \frac{m}{m_1} = \frac{117}{9} \Rightarrow 13$$

$$M_2 = \frac{m}{m_2} = \frac{117}{13} \Rightarrow 9$$

Now,

$$m_1^{-1} = 7$$

$$m_2^{-1} = 3$$

$$x = (a_1 m_1 m_1^{-1} + a_2 m_2 m_2^{-1}) \pmod{m}$$

$$x = (3 \times 13 \times 7 + 7 \times 9 \times 3) \pmod{117}$$
~~$$x = (3 \times 13 \times 7 + 7 \times 9 \times 3) \pmod{117}$$~~

$$x = 462 \pmod{117}$$

$$x = 111$$

5.

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

$$x = [a_1 m_1 m_1^{-1} + a_2 m_2 m_2^{-1} + a_3 m_3 m_3^{-1}] \pmod{M}$$

$x \equiv a_1 \pmod{m_1}$	$x \equiv 2 \pmod{3}$
$x \equiv a_2 \pmod{m_2}$	$x \equiv 3 \pmod{5}$
$x \equiv a_3 \pmod{m_3}$	$x \equiv 2 \pmod{7}$

~~Ques~~

Given		to find		
$a_1 = 2$	$m_1 = 3$	m_1	m_1^{-1}	M
$a_2 = 3$	$m_2 = 5$	m_2	m_2^{-1}	
$a_3 = 2$	$m_3 = 7$	m_3	m_3^{-1}	

Note: Here 3, 5 & 7 are relatively prime.

$\therefore m = m_1 \times m_2 \times m_3$
 $m = 3 \times 5 \times 7$
 $m = 105$

Now, $m_1 = \frac{m}{m_1} = \frac{105}{3} \Rightarrow 35$
 $m_2 = \frac{m}{m_2} = \frac{105}{5} \Rightarrow 21$
 $m_3 = \frac{m}{m_3} = \frac{105}{7} \Rightarrow 15$

Now,
 $m_1 \times m_1^{-1} = 1 \pmod{m_1}$
 $35 \times m_1^{-1} = 1 \pmod{3}$
 $35 \times 2 = 1 \pmod{3}$
 $\boxed{m_1^{-1} = 2}$

$$\rightarrow m_2 \times m_2^{-1} = 1 \pmod{m_2}$$

$$21 \times m_2^{-1} = 1 \pmod{5}$$

$$21 \times 1 = 1 \pmod{5}$$

$$\boxed{m_2^{-1} = 1}$$

$$\rightarrow m_3 \times m_3^{-1} = 1 \pmod{m_3}$$

$$15 \times m_3^{-1} = 1 \pmod{7}$$

$$15 \times 1 = 1 \pmod{7}$$

$$\boxed{m_3^{-1} = 1}$$

Now

$$x = (a_1 m_1 m_1^{-1} + a_2 m_2 m_2^{-1} + a_3 m_3 m_3^{-1}) \pmod{M}$$

$$= (2 \times 35 \times 2 + 3 \times 21 \times 1 + 2 \times 15 \times 1) \pmod{105}$$

$$= 233 \pmod{105}$$

$$\boxed{x = 23} \quad \checkmark$$